Lecture 4: More on Integration

20 August 2025

Exercise. Use the substitution $x = \tan u$ to evaluate

$$\int \frac{1}{1+x^2}\,dx.$$

Hence, or otherwise, determine

$$\int \frac{1}{a^2+x^2}\,dx.$$

Solution. Let $x = \tan u$, with $dx = \sec^2 u\,du$. Then, the integral becomes

$$\int \frac{1}{1+\tan^2 u}\,\sec^2 u\,du = \int \frac{\sec^2 u}{\sec^2 u}\,du = \int 1\,du = u + c.$$

Since $u = \tan^{-1} x$, we find that

$$\int \frac{1}{1+x^2}\,dx = \tan^{-1} x + c.$$

Now, observe that

$$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a^2}\int \frac{1}{1+(x/a)^2}\,dx.$$

We can evaluate this with a substitution: let $u = x/a$, with $du = (1/a)\ dx$. Thus,

$$\frac{1}{a^2}\int \frac{1}{1+(x/a)^2}\,dx = \frac{1}{a^2}\int \frac{a}{1+u^2}\,du = \frac{1}{a}\tan^{-1} u + c = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c.$$

In general, there is no set algorithm to evaluate a given integral—in fact, very few integrals can be evaluated explicitly. We introduce a few more integration techniques for specific types of integrands, after which we examine some applications of integration to measurement.

Integration by Partial Fractions

A *rational function* is a ratio of polynomials:

$$f(x) = \frac{p(x)}{q(x)} = \frac{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0}.$$

In the above expression, the numbers n and m are called the *degrees* of the polynomials p and q, respectively. As it turns out, all rational functions can be integrated: this can be done by decomposing a given rational function into a sum of partial fractions. The key idea behind partial fractions relies on the *Fundamental Theorem of Algebra*:

Theorem (Fundamental Theorem of Algebra)**.** *Every polynomial with real coefficients can be factored into a product of linear and irreducible quadratic factors* with real coefficients.*

* A linear factor is of the form $ax + b$, while an irreducible quadratic is of the form $ax^2 + bx + c$, where $ax^2 + bx + c = 0$ has no real solutions and so cannot be further factorized.

The Fundamental Theorem of Algebra is quite a powerful statement: this means that we will always be able to factorize the denominator of a rational function, and we can thus rewrite every rational function as a sum of fractions whose denominators are either constants, linear polynomials, or quadratic polynomials.

Example. Let us evaluate

$$\int \frac{x^3 - x^2 + 2x + 2}{x^2 + 1}\, dx.$$

To begin, observe that the *degree* of the numerator is larger than that of the denominator, so we can simplify this expression by long division:

$$\begin{array}{r|rrrr}
 & & & x & -1 \\ \hline
x^2+1 & x^3 & -x^2 & +2x & +2 \\
 & -x^3 & & -x & \\ \hline
 & & -x^2 & +x & +2 \\
 & & x^2 & & +1 \\ \hline
 & & & x & +3
\end{array}$$

Thus, we have

$$\int \frac{x^3 - x^2 + 2x + 2}{x^2 + 1}\, dx = \int x - 1 + \frac{x+3}{x^2+1}\, dx.$$

We can now proceed to integrate term by term:

$$\int x - 1 + \frac{x}{x^2+1} + \frac{3}{x^2+1}\, dx = \frac{x^2}{2} - x + \frac{1}{2}\ln\left(x^2+1\right) + 3\tan^{-1} x + c.$$

Example. Let us evaluate

$$\int \frac{2x^2 - x + 4}{x^3 + 4x}\, dx.$$

The denominator factors into

$$x^3 + 4x = x\left(x^2+4\right),$$

and so we have a partial fraction decomposition

$$\frac{A}{x} + \frac{Bx + C}{x^2+4} = \frac{A\left(x^2+4\right) + (Bx+C)\,x}{x\left(x^2+4\right)} = \frac{(A+B)\,x^2 + Cx + 4A}{x^3+4x}.$$

Comparing coefficients, we require that

$$A + B = 2, \quad C = -1, \quad 4A = 4.$$

Thus, $A = 1$, $B = 1$, and $C = -1$ and so

$$\begin{aligned}
\int \frac{2x^2 - x + 4}{x^3 + 4x}\, dx &= \int \frac{1}{x} + \frac{x-1}{x^2+4}\, dx \\
&= \int \frac{1}{x}\, dx + \int \frac{x}{x^2+4}\, dx - \int \frac{1}{x^2+4}\, dx \\
&= \ln|x| + \frac{1}{2}\ln\left(x^2+4\right) - \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + c.
\end{aligned}$$

We summarize the different possible partial fraction decompositions below. For the sake of simplicity, let us assume[†] that the rational function to be integrated is **proper**:

$$f(x) = \frac{p(x)}{q(x)}, \text{ where } \deg(p) < \deg(q).$$

By the Fundamental Theorem of Algebra, we may factor q into linear and quadratic factors, and we can thus express $f(x)$ as a sum of partial fractions of the form

$$\frac{A}{(ax+b)^i} \quad \text{or} \quad \frac{Ax+B}{(ax^2+bx+c)^j}.$$

[†] Note that this is a valid assumption since, if we were to start with an *improper* rational function, we can always perform long division to express it as a sum of a polynomial and a proper function:

$$f(x) = \frac{p(x)}{q(x)} = s(x) + \frac{r(x)}{q(x)},$$

where $\deg(r) < \deg(q)$.

1. If $q(x)$ is a product of *distinct* linear factors
$$q(x) = (a_1x + b_1)(a_2x + b_2)\cdots(a_nx + b_n),$$
then
$$\frac{p(x)}{q(x)} = \frac{A_1}{a_1x + b_1} + \frac{A_2}{a_2x + b_2} + \cdots + \frac{A_n}{a_nx + b_n}.$$
2. If the denominator contains a *repeated* linear factor $(ax + b)^k$, then the partial fraction decomposition must contain
$$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \cdots + \frac{A_k}{(ax + b)^k}.$$
3. If the complete factorization of the denominator contains an irreducible quadratic factor $ax^2 + bx + c$, then the partial fraction decomposition will have a term of the form
$$\frac{Ax + B}{ax^2 + bx + c}.$$

Example. Let us find
$$\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1}\,dx.$$
Observe that the degree of the numerator is larger than that of the denominator; let us first perform long division.
$$\begin{array}{r|rrrrr}
 & & & & & x+1 \\ \hline
x^3 - x^2 - x + 1 & x^4 & & -2x^2 & +4x & +1 \\
 & -x^4 & +x^3 & +x^2 & -x & \\ \hline
 & & x^3 & -x^2 & +3x & +1 \\
 & & -x^3 & +x^2 & +x & -1 \\ \hline
 & & & & 4x &
\end{array}$$
Thus, we have
$$\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1}\,dx = \int x + 1 + \frac{4x}{x^3 - x^2 - x + 1}\,dx.$$
To evaluate the final term of the integral, we factor the denominator:
$$x^3 - x^2 - x + 1 = (x + 1)(x - 1)^2.$$
Thus, we have a partial fraction decomposition
$$\begin{aligned}
\frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2} &= \frac{A(x-1)^2 + B(x+1)(x-1) + C(x+1)}{(x+1)(x-1)^2} \\
&= \frac{(A+B)x^2 + (-2A+C)x + A - B + C}{x^3 - x^2 - x + 1}.
\end{aligned}$$
Comparing coefficients, we have
$$A + B = 0, \quad -2A + C = 4, \quad A - B + C = 0.$$
We have $B = -A$ and $C = 4 + 2A$, and so
$$A - B + C = A - (-A) + (4 + 2A) = 4A + 4 = 0.$$
Thus, $A = -1$, $B = 1$, and $C = 2$, and so
$$\begin{aligned}
\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1}\,dx &= \int x + 1 - \frac{1}{x+1} + \frac{1}{x-1} + \frac{2}{(x-1)^2}\,dx \\
&= \frac{x^2}{2} + x - \ln|x+1| + \ln|x-1| - \frac{2}{x-1} + c.
\end{aligned}$$

Improper Integrals

We briefly revisit the problem of evaluating the definite integral

$$\int_a^t f(x)\ dx.$$

If this expression is defined for every $t \geq a$, we can define the **improper integral**

$$\int_a^\infty f(x)\ dx = \lim_{t\to\infty} \int_a^t f(x)\ dx.$$

We can similarly consider the definite integral

$$\int_t^b f(x)\ dx.$$

If the integral exists for all $t \leq b$, we define

$$\int_{-\infty}^b f(x)\ dx = \lim_{t\to-\infty} \int_t^b f(x)\ dx.$$

If these limits exists, then we say that the improper integral is **convergent**; otherwise, the integral is said to be **divergent**. Note that, if both integrals are convergent, then we may define

$$\int_{-\infty}^\infty f(x)\ dx = \int_{-\infty}^a f(x)\ dx + \int_a^\infty f(x)\ dx.$$

Geometrically, this suggests that some regions that extend to infinity might in fact have finite areas.

Example. Let us consider the improper integral

$$\int_1^\infty \frac{1}{x^p}\ dx.$$

We might be interested in determining the conditions on p for which the above integral converges. We consider some cases.

- When $p = 1$, we can directly evaluate

  $$\lim_{t\to\infty} \int_1^t \frac{1}{x}\ dx = \lim_{t\to\infty} [\ln t - \ln 1] = \lim_{t\to\infty} \ln t \to \infty,$$

  and so the integral diverges.

- When $p \neq 1$, we have

  $$\begin{aligned}\lim_{t\to\infty} \int_1^t x^{-p}\ dx &= \lim_{t\to\infty} \left[\frac{x^{-p+1}}{-p+1} \bigg|_{x=1}^{x=t} \right] \\ &= \lim_{t\to\infty} \frac{t^{1-p} - 1}{1-p}.\end{aligned}$$

  Observe that as $t \to \infty$,

  - $t^{1-p} \to 0$ when $p > 1$, and the integral converges, while
  - $t^{-1-p} \to \infty$ when $p < 1$, and the integral diverges.

Hence, the integral is divergent when $p \leq 1$, and it is convergent when $p > 1$, with

$$\int_1^\infty \frac{1}{x^p}\ dx = \frac{1}{p-1}.$$

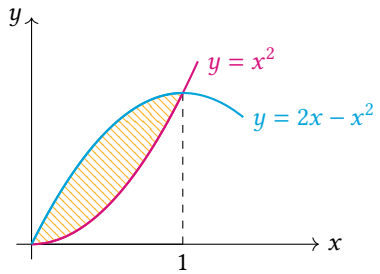
Measurement

By definition, the definite integral

$$\int_a^b f(x)\ dx$$

computes the *signed area* between the graph of $f(x)$ and the x-axis.

Example. Suppose we wish to find the area enclosed by the graphs of $y = x^2$ and $y = 2x - x^2$.



Note that the area of the shaded region can be obtained by taking the difference of the areas beneath $y = 2x - x^2$ and $y = x^2$:

$$\begin{aligned}\text{area} &= \int_0^1 2x - x^2\ dx - \int_0^1 x^2\ dx = \int_0^1 \left(2x - x^2\right) - \left(x^2\right)\ dx \\ &= \int_0^1 2x - 2x^2\ dx = x^2 - \frac{2}{3}x^3\Big|_{x=0}^{x=1} = 1 - \frac{2}{3} = \frac{1}{3}.\end{aligned}$$

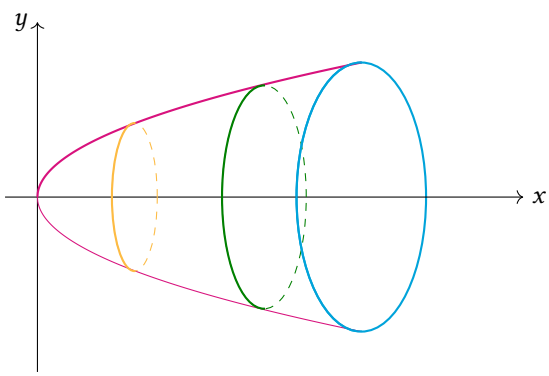
Integration is thus a useful tool in *measurement*: computing areas, as in the example above, and even in computing volumes and lengths.

We first consider the problem of computing *volumes*: for a solid with a constant cross-section, we know that

$$\text{volume} = (\text{area of base}) \times (\text{height})\,.$$

For a solid with a varying cross-section, we can compute the volume by 'adding' the areas of each of the infinitely many cross sections.

Example. Consider the graph of $y = \sqrt{x}$ from $x = 0$ to $x = 1$. We can form a *solid of revolution* by rotating this graph about the x-axis.



Observe that the cross-sections of the solid are all circles whose radii vary as a function of x.

- When $x = 0.3$, the radius is $r = \sqrt{0.3}$;
- when $x = 0.7$, the radius is $r = \sqrt{0.7}$;
- when $x = 0.1$, the radius is $r = 1$.

In particular, for each value of x between 0 and 1, the cross-section has radius $y = \sqrt{x}$, and so the area of each cross-section is given by

$$A(x) = \pi r^2 = \pi \left(\sqrt{x}\right)^2 = \pi x.$$

We can thus compute the volume of the entire solid by 'adding' all of these areas: that is, by *integrating* the area function from $x = 0$ to $x = 1$:

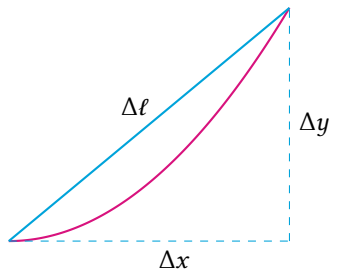
$$\text{volume} = \int_0^1 A(x)\ dx = \int_0^1 \pi x\, dx = \pi \left.\frac{x^2}{2}\right|_{x=0}^1 = \frac{\pi}{2}.$$

More generally, given a solid of revolution generated by rotating the graph of a function $y = f(x)$, from $x = a$ to $x = b$, about the x-axis, its volume is given by the definite integral

$$\int_a^b \pi (f(x))^2\ dx.$$

In the above expression, the integrand $\pi (f(x))^2$ gives the *area* of a cross-section, which is a circle of radius $r = f(x)$.

We can similarly use integrals to compute *arc length*. Consider a smooth function $y = f(x)$; suppose we wish to compute an infinitesimally small length $\Delta\ell$ along this curve.



Observe that

$$\Delta\ell = \sqrt{(\Delta x)^2 + (\Delta y)^2} = \Delta x \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2} \quad \Longrightarrow \quad \frac{\Delta\ell}{\Delta x} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2}.$$

In particular, we have an expression that allows us to compute the arc length of a function $y = f(x)$ from $x = a$ to $x = b$:

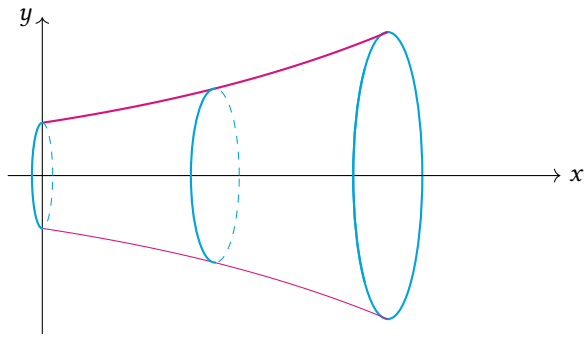
$$\frac{d\ell}{dx} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2} \quad \Longrightarrow \quad \ell = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$$

The arc length formula allows us to compute the *surface area* of a solid of revolution. Given a solid formed by rotating the curve $y = f(x)$, from $x = a$ to $x = b$, about the x-axis, the area of its surface is given by the definite integral

$$\int_a^b (2\pi f(x)) \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$$

Here, the factor of $2\pi f(x)$ gives the *circumference* of a cross-section of the solid.

Example. Let us compute the area of the surface generated by rotating the curve $y = e^x$, from $x = 0$ to $x = 1$, about the x-axis.



Note that

$$\frac{dy}{dx} = \frac{d}{dx}e^x = e^x.$$

Hence, the surface area is given by

$$\int_0^1 (2\pi e^x)\sqrt{1 + (e^x)^2}\,dx.$$

We evaluate this integral with a substitution: let $u = e^x$, with $du = e^x\,dx$. Then, the bounds of integration go from $u = e^0 = 1$ to $u = e^1 = e$, and

$$\int_0^1 (2\pi e^x)\sqrt{1 + (e^x)^2}\,dx = 2\pi\int_1^e \sqrt{1 + u^2}\,du.$$

Evaluating this integral will yield the desired surface area.